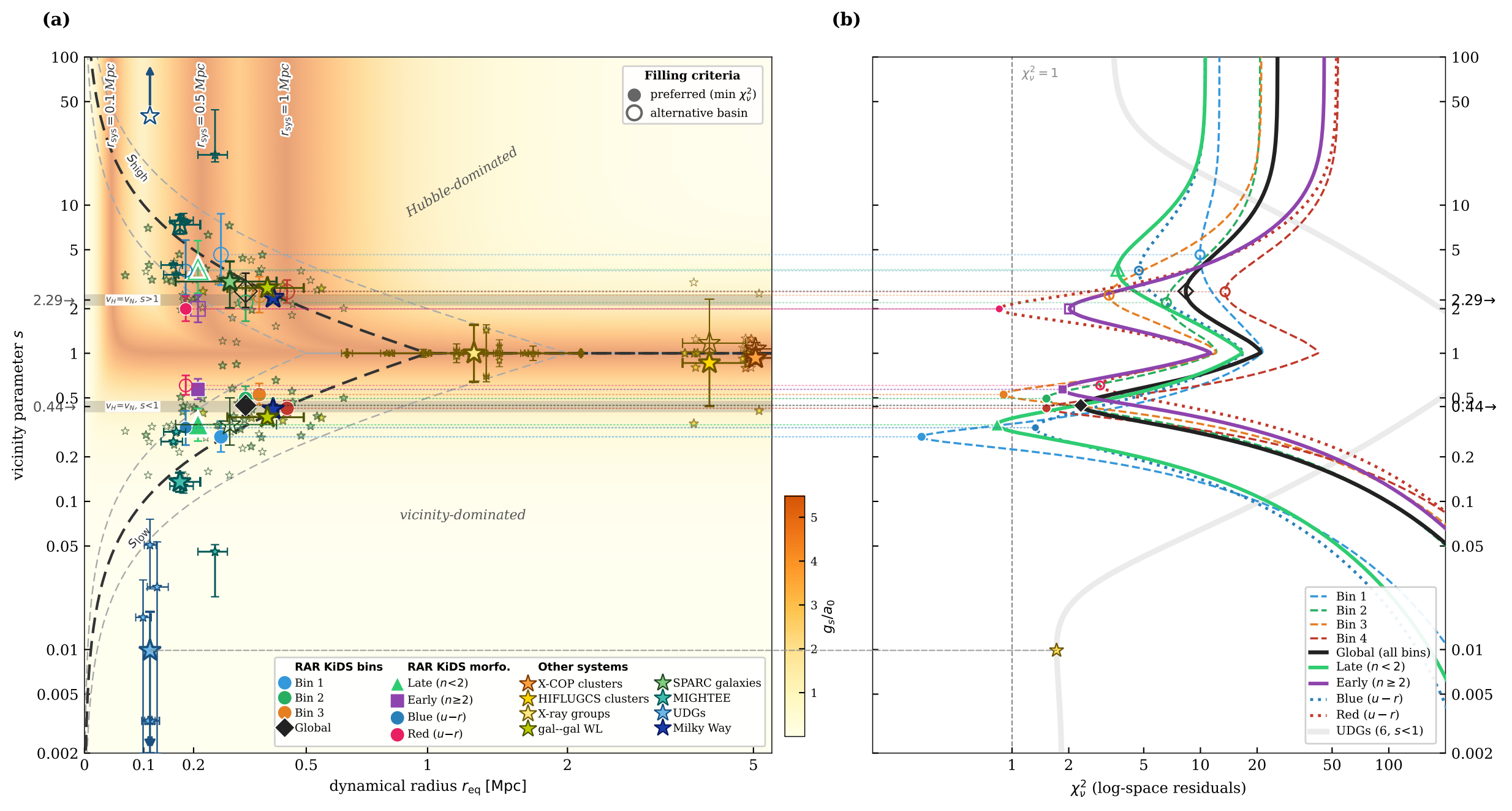

(a)
(b)
Filling criteria
preferred (min χ^2_ν)
alternative basin
$r_{sys} = 0.1$ Mpc
$r_{sys} = 0.5$ Mpc
$r_{sys} = 1$ Mpc
s_{high}
s_{low}
Hubble-dominated
vicinity-dominated
$v_H = v_N$, $s>1$
$v_H = v_N$, $s<1$
2.29→
0.44→
vicinity parameter s
dynamical radius r_{eq} [Mpc]
g_s/a_0
RAR KiDS bins
Bin 1
Bin 2
Bin 3
Global
RAR KiDS morfo.
Late (n<2)
Early (n≥2)
Blue ($u-r$)
Red ($u-r$)
Other systems
X-COP clusters
HIFLUGCS clusters
X-ray groups
gal--gal WL
SPARC galaxies
MIGHTEE
UDGs
Milky Way
$\chi^2_\nu = 1$
χ^2_ν (log-space residuals)
Bin 1
Bin 2
Bin 3
Bin 4
Global (all bins)
Late ($n < 2$)
Early ($n \geq 2$)
Blue ($u-r$)
Red ($u-r$)
UDGs (6, s<1)